MATERIALS

ITEM	PART	SECTION / MATERIAL	CUT LEN	QTY	QTY x10	OPERATIONS	kg EA
1	Pallet foot	Pressed galv 130x130x50, 20 hole	-	4	40	Purchased. Weld flange-up under item 2 (see Detail B).	0.59
2	Corner stub	50x50x4 SHS C350L0	250	4	40	Square cut. Drill 13 dia thru at 25 from top (one axis).	1.33
3	Telescopic post	40x40x3 SHS C350L0	1530	4	40	Square cut. Drill 13 dia thru at 225 from bottom. 8 dia vent at 20 from top.	5.05
4	Post cap	5 PL 250 grade	40x40	4	40	Laser cut, 16.5 dia centre hole. Weld to post top.	0.05
5	Locating pin	16 dia round bar	25	4	40	Saw cut, chamfer 1x45 top. Plug weld into item 4, 20 proud.	0.04
6	Side rail	50x50x2 SHS C350L0	1900	2	20	Square cut both ends.	5.57
7	End bearer	50x50x2 SHS C350L0	1600	2	20	Square cut both ends.	4.69
8	Centre rail	50x50x3 SHS C350L0	1900	1	10	Square cut both ends. Top flush with items 6/7.	8.07
9	Corner gusset	4 PL 250 grade	80x80	4	40	Laser cut right triangle, 15 snip. Weld to underside of rail corners.	0.10
10	Retainer bolt	M12 x 70 grade 8.8 galv + nylok	-	4	40	Fit through items 2 and 3 after assembly.	-

STOCK ORDER (8.0 m lengths)

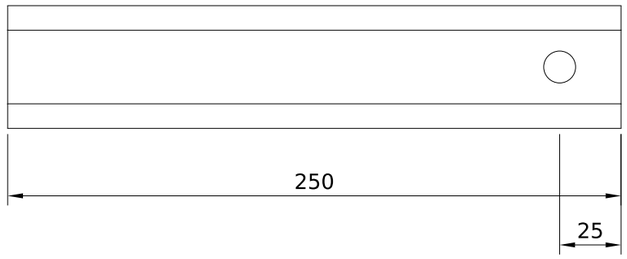
	PER RACK	PER 10 RACKS
50x50x4 SHS (item 2)	1.00 m	2 lengths (32 x 250 / length)
40x40x3 SHS (item 3)	6.12 m	8 lengths (5 x 1530 / length)
50x50x2 SHS (items 6, 7)	7.00 m	10 lengths (1 per rack: 2 x 1900 + 2 x 1600)
50x50x3 SHS (item 8)	1.90 m	3 lengths (4 x 1900 / length)

FABRICATION NOTES

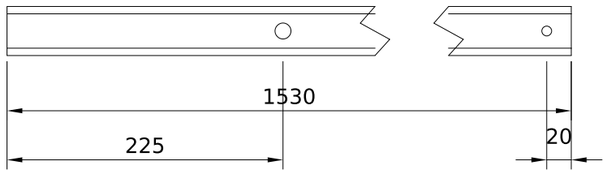
1. Material: SHS/RHS C350L0 to AS/NZS 1163. Plate 250 grade to AS/NZS 3678.
2. Weld: 3 mm fillet all round unless noted. 2.5 mm to 2 mm wall. GMAW.
3. Rails 6, 7, 8 all 50 deep, flush top 200 / bottom 150. Welded between stubs.
4. Post 3 slides in stub 2 and bears on foot 1. Check post slides freely after galv.
5. Hot dip galvanise after fab. Vent holes per galvaniser, max 8 dia visible.
6. Gussets 9 on underside so box seat stays flat. Tolerance +/-2 overall.
7. Forklift: forks min 1200 apart. One rack at a time. Check capacity plate.

PART DETAILS (NTS, dims mm)

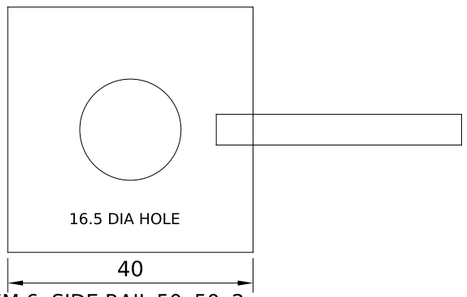
ITEM 2 50x50x4



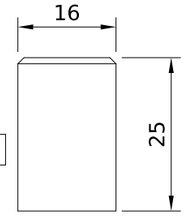
ITEM 3 40x40x3



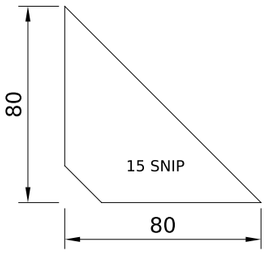
ITEM 4 5 PL



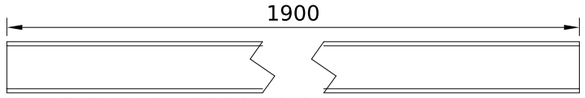
ITEM 5 16 RD



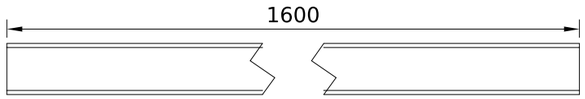
ITEM 9 4 PL (1:2.5)



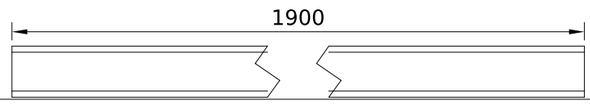
ITEM 6 SIDE RAIL 50x50x2

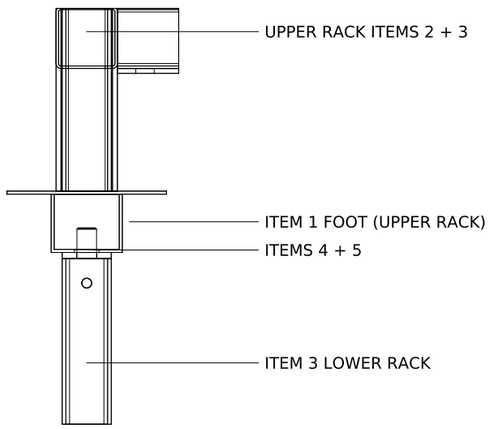


ITEM 7 END BEARER 50x50x2

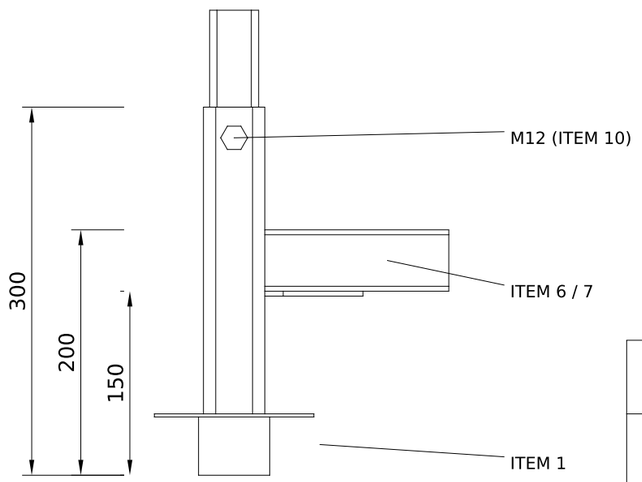


ITEM 8 CENTRE RAIL 50x50x3





DETAIL A STACKING JOINT 1:5



DETAIL B CORNER 1:5

RAVEN INDUSTRIES PTY LTD	DWG RI-RACK-001 REV F
UMBRELLA BOX STACKING RACK 15 BOX CUT LIST AND PART DETAILS	SHEET 2 OF 2 A3 DATE 11/11/2026 mm

PROOF OF CONCEPT: GENERATED FROM THE 3D MODEL (rack_model.py), NOT HAND DRAWN